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Remark AS R78: A Remark on Algorithm AS 178: The Gauss-Jordan Sweep Operator with Detection of Collinearity

Author(s): Martin S. Ridout and John M. Cobby

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```
      Cliques[NoOfCliques] := ThisClique - [n];
      END
    ELSE
     WITH Graph DO
      BEGIN
       NoOfCliques := NoOfCliques + 1;
       Cliques[NoOfCliques] := ThisClique
      END
  END;

   { 2. Compare each clique in Subgraph with those directly placed in
     Graph.  If it is a subset of any clique in Graph it is ignored,
     otherwise it is added to Graph. The searching process uses a
     variation of the technique of adding the item as a sentinel at the
     end of the sequence being searched to simplify the loop termination.
     Here, since Graph is being extended but the searching takes place
     only over its initial part, a buffer is left at the interface to be
     filled in just before exit.  }

 WITH Graph DO
  BEGIN
   NoOfCliques := NoOfCliques + 1;
   buffer := NoOfCliques;
   FOR i := 1 TO Subgraph.NoOfCliques DO
    BEGIN
     ThisClique := Subgraph.Cliques[i];
     Cliques[buffer] := ThisClique;
     j := 1;
     WHILE NOT (ThisClique <= Cliques[j]) DO
      j := j + 1;
     IF j = buffer THEN
      BEGIN
       NoOfCliques := NoOfCliques + 1;
       Cliques[NoOfCliques] := ThisClique
      END
    END;
   Cliques[buffer] := Cliques[NoOfCliques];
   NoOfCliques := NoOfCliques - 1
  END
END;{of DeleteEdge }
```

Remark AS R78

A Remark on Algorithm AS 178: The Gauss–Jordan Sweep Operator with Detection of Collinearity

By Martin S. Ridout† and John M. Cobby

Animal and Grassland Research Institute, Maidenhead, UK

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Algorithm AS 178 (Clarke, 1982) can encounter problems caused by rounding errors. The coding depends on the fact that, in theory, the sign of T(KK) determines the parity of the number of previous pivots on variable K:

T(KK) .LT. ZERO	if there has been an odd number of previous pivots;
T(KK) .GT. ZERO	if there has been an even number (including zero) of previous pivots.

† *Address for correspondence*: Institute of Horticultural Research, East Malling, Maidstone, Kent, ME19 6BJ, UK.

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In practice, however, the sign of T(KK) may be incorrect, owing to rounding error. A simple remedy is to use a separate logical array SWEPT as an additional argument to the subroutine and to arrange that

SWEPT(K) is .TRUE. if there has been an odd number of previous pivots on variable *K* or

SWEPT(K) is .FALSE. if there has been an even number of previous pivots on variable *K*.

The changes detailed here also cause the algorithm to check for consistency between the values of the array SWEPT and the signs of the diagonal elements of *T*, whenever a collinearity check is made. If an inconsistency is detected the subroutine terminates without performing a pivot and with IFAULT set to 2. The parameter *L* indicates the first variable for which the consistency check failed.

Small diagonal elements of the wrong sign may, perhaps, be ignored, but larger elements indicate the possibility of severe numerical problems. In any event, the action to be taken when rounding error is detected is left to the discretion of the calling program.

A referee has pointed out that, if the modified algorithm terminates with IFAULT = 2, this may be because the matrix *T* is negative definite. However, *T* is intended to be a sums of squares and products matrix. Thus, provided that a numerically stable algorithm is used to form the matrix *T* (e.g. algorithm AS 12, Herraman (1968)), negative definiteness is likely to arise only for highly collinear data.

The following changes are required.

(a) Modify the structure of the subroutine as follows:
SUBROUTINE GSWEEP(S, T, K, L, M, N, E, SWEPT, IFAULT)
Additional formal parameter
SWEPT Logical array (*N*) input: before the first call to GSWEEP all elements should be set to .FALSE. Thereafter the contents of this array should not be altered between successive calls to the subroutine

(b) Immediately above the DATA statement, insert the declaration

```
LOGICAL SWEPT(N)
```

(c) Ten lines above the statement labelled 10 replace

```
IF (T(KK) .LT. ZERO) GOTO 20
IF (T(KK) .LT. E * S(K)) RETURN
```

by

```
IF (SWEPT(K) .AND. T(KK) .LT. ZERO) GOTO 20
IF (SWEPT(K) .AND. T(KK) .GT. ZERO) GOTO 95
IF (T(KK) .LT. ZERO) GOTO 95
IF (T(KK) .LE. E * S(K)) RETURN
```

(d) Two lines above the statement labelled 10 replace

```
IF (T(II) .GE. ZERO) GOTO 10
```

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by

```
IF (.NOT. SWEPT(L) .AND. T(II) .GE. ZERO) GOTO 10
IF (.NOT. SWEPT(L)) GOTO 95
IF (T(II) .GT. ZERO) GOTO 95
```

(e) Insert the following three statements immediately after the statement labelled 90:

```
      SWEPT(K) = .NOT. SWEPT(K)
      RETURN
   95 IFAULT = 2
```

Example

The matrix

$$\begin{pmatrix} 30 & 16 & 46 \\ 16 & 10 & 26 \\ 46 & 26 & 72 \end{pmatrix}$$

is singular (row 3 = row 1 + row 2), but a single-precision version of algorithm AS 178 running on a VAX 11/750 computer allows successive pivots on rows 1, 2 and 3 without detecting collinearity. This is because after pivoting on rows 1 and 2 element (3, 3) of the matrix is -4.77×10^{-6}. Since this is a negative value, algorithm AS 178 incorrectly deduces that there has been an odd number of previous pivots on row 3 and therefore does not perform collinearity checks.

The revised algorithm returns without performing the pivot on row 3 and with IFAULT set to 2 and L set to 3 indicating the inconsistency in the sign of element (3, 3) of the matrix.

Acknowledgement

We are grateful to a referee for helpful suggestions.

References

Clarke, M. R. B. (1982) Algorithm AS 178: The Gauss–Jordan sweep operator with detection of collinearity. *Appl. Statist.*, **31**, 166–168.

Herraman, C. (1968) Algorithm AS 12: Sums of squares and products matrix. *Appl. Statist.*, **17**, 289–292.

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